# HEART DISEASE PREDICTION REPORT

Explainable Deep Learning Diagnosis Platform (CNN-LSTM-Attention + Grad-CAM)

| | | | |
|---|---|---|---|
| **Patient Index** | #4 | **True Label (Ground Truth)** | Normal |
| **Model Prediction Probability** | 29.81% | **Analysis Date** | 2026-06-23 |

## Model Recommendation: NORMAL / LOW RISK

## Patient Clinical Profile

| | | | |
|---|---|---|---|
| **Age** | 54 years | **Sex** | Female |
| **Chest Pain Type (CP)** | Non-anginal Pain | **Resting Blood Pressure** | 150 mm Hg |
| **Serum Cholesterol** | 232 mg/dl | **Fasting Blood Sugar (FBS)** | Normal (<=120 mg/dl) |
| **Resting ECG** | Left Ventricular Hypertrophy | **Max Heart Rate (Thalach)** | 165 bpm |
| **Exercise Induced Angina** | No | **ST Depression (Oldpeak)** | 1.6 |
| **ST Slope** | Upsloping | **Vessels via Fluoroscopy (CA)** | 0 |
| **Thalassemia (Thal)** | Reversible Defect | **Predicted Diagnosis Probability** | **29.81%** |

## Primary Diagnostic Drivers (AI Focus)

### 1. CP

**Patient Value:** 3.0 | **Ideal Range:** 1, 2, or 3 (Angina/Non-anginal pain)

***Interpretation:*** *Chest pain type 3 (Angina or localized chest pain).*

### 2. CA

**Patient Value:** 0.0 | **Ideal Range:** 0 vessels colored

***Interpretation:*** *0 major vessels colored (clear arterial paths).*

### 3. RESTECG

**Patient Value:** 2.0 | **Ideal Range:** 0 (Normal)

***Interpretation:*** *Abnormal ECG resting results (type 2), suggesting hypertrophy or ST-T wave anomalies.*

### 4. THAL

**Patient Value:** 7.0 | **Ideal Range:** 3 (Normal)

***Interpretation:*** *Reversible defect thalassemia observed (indicating active myocardial ischemia under stress).*

## Secondary Clinical Abnormalities (Outside Top AI Focus)

### 1. TRESTBPS

**Patient Value:** 150.0 | **Ideal Range:** <120 mmHg

***Interpretation:*** *Measured at 150.0 mmHg, indicating Stage 2 Hypertension which significantly strains cardiac walls.*

### 2. CHOL

**Patient Value:** 232.0 | **Ideal Range:** <200 mg/dl

***Interpretation:*** *Measured at 232.0 mg/dl, which is borderline high.*

### 3. OLDPEAK

**Patient Value:** 1.6 | **Ideal Range:** 0.0 mm (No ST deviation)

***Interpretation:*** *Measured at 1.6 mm. Significant ST segment depression suggests active myocardial ischemia during stress.*

## Explainability Visualization (Attention vs. Grad-CAM)

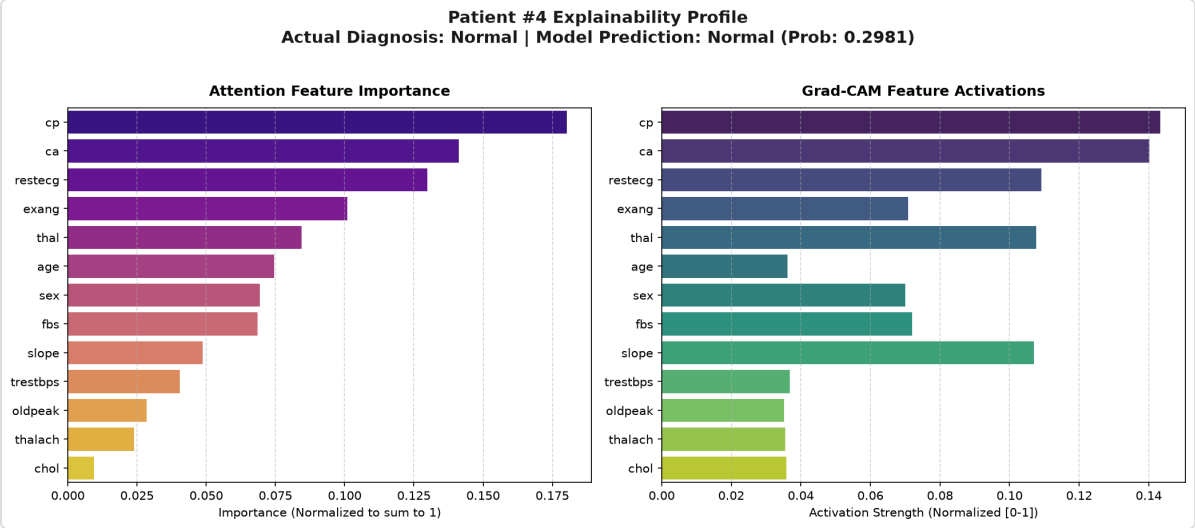



Figure 1: Dual-Perspective feature attribution showing global attention weight and patient-specific Grad-CAM contribution.

## 2D Gramian Angular Field (GASF) Heatmap

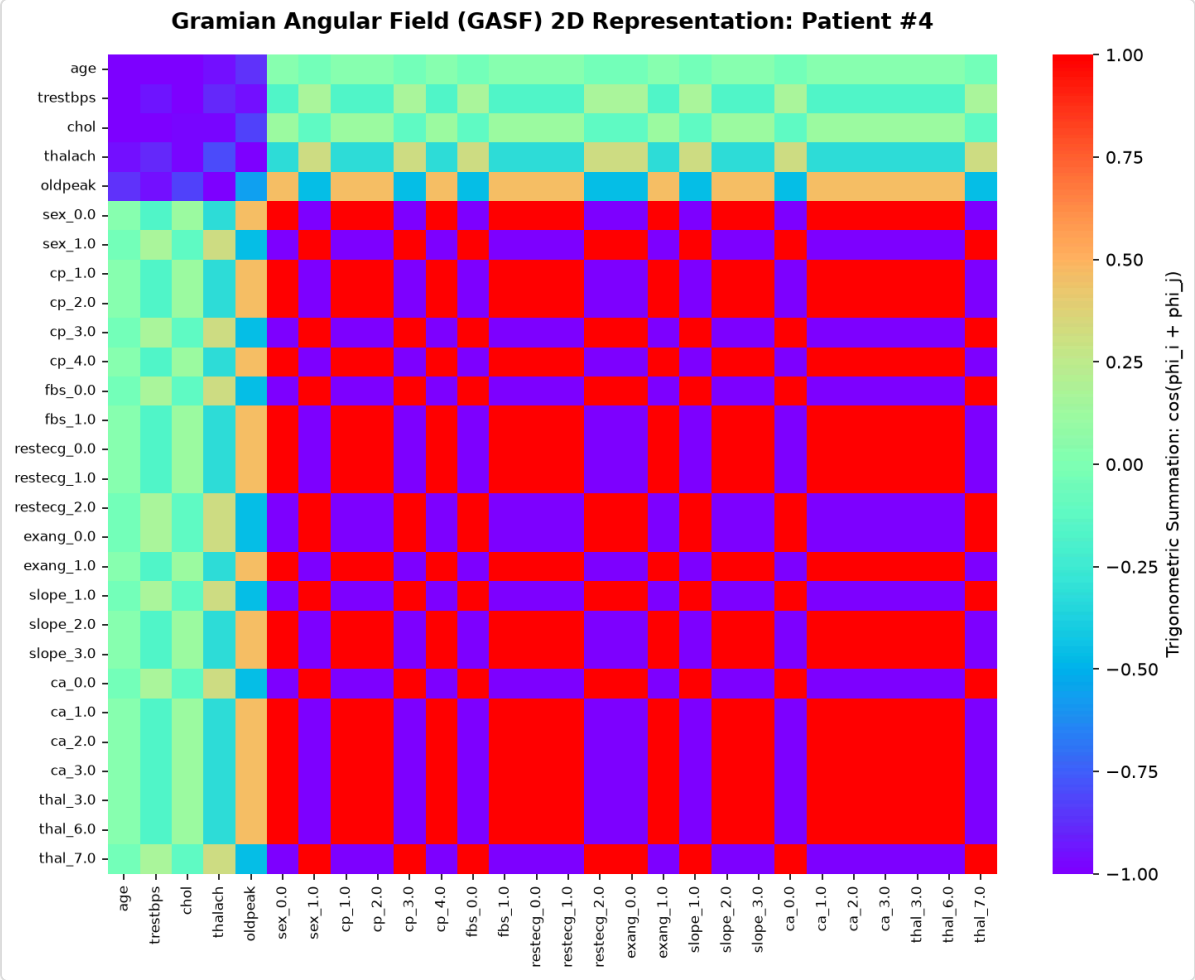



Figure 2: 2D feature-feature interaction texture mapping derived via Gramian Angular Summation Field conversion.